TreeLoom: Learning Local Ancestry from Complete Genealogies

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Abstract

Local ancestry inference (LAI) assigns genomic segments to ancestral source populations. Genealogies can inform these assignments by showing how a query genome is related to reference genomes, but recombination changes those relationships along a chromosome. We introduce TreeLoom, a neural model that processes every node of an inferred local tree and follows the same sampled haplotypes through the ordered trees. Messages moving toward and away from the root combine evidence across each genealogy. Structural features describe changes between neighboring trees, sample memory carries the history of each sampled haplotype, and a final recurrent network combines genomic context before ancestry prediction. On eight reused European development simulations, TreeLoom reaches $97.08 \pm 0.57\%$ span accuracy for modern queries, compared with $94.45 \pm 1.15\%$ for the evaluated ARGMix release and $93.78 \pm 0.84\%$ for a derived four source AncestralPaths endpoint. These are comparisons of fitted pipelines with different inputs and training, rather than exact publication reproductions. A matched control without added sample memory reaches $97.00 \pm 0.50\%$ and has higher boundary F1. The results motivate learning from complete genealogies while showing that accurate genomic assignment and accurate tract boundaries remain distinct objectives.

1 Introduction

Genomes from admixed populations contain segments inherited from different ancestral sources. Local ancestry inference (LAI) seeks to locate these segments and identify their sources. The resulting ancestry tracts can reveal when populations mixed and how genetic material moved between populations. Ancient genomes add a temporal dimension by sampling individuals who lived before, during and after major changes in ancestry. Their interpretation depends on both the relationships among sampled genomes and the continuity of ancestry along a chromosome.

An inferred genealogy offers a direct representation of those relationships. At a genomic position, a local tree connects sampled haplotypes through common ancestors. A query and a reference genome may share a recent branch in one region and a more distant ancestor in another. An ancestral recombination graph (ARG) represents this changing history, and its local trees provide a sequence of related prediction problems. Evidence from the whole tree can help distinguish ancestral sources, while evidence from neighboring regions can help stabilize local assignments.

Existing methods use this structure in different ways. AncestralPaths classifies genealogical paths in analyses of ancient Eurasian ancestry (Irving-Pease et al. 2024). ARGMix combines graph representations with genomic context (Shanks et al. 2026). ARGformer learns representations of genealogical paths and supports ancestry assignment through reference retrieval (Bonet et al. 2026). These methods establish the value of genealogies for LAI and motivate a question about representation: can a compact model retain the complete local genealogy while maintaining a coherent representation of each sampled haplotype along the chromosome?

This question has a biological constraint. Sample identities persist across local trees, but inferred internal ancestors need not have a reliable correspondence. Reusing an identifier assigned to an inferred ancestor can therefore connect unrelated states. A model also needs to distinguish a change in tree structure from a change in ancestry. Recombination alters genealogy, yet many neighboring trees can still support the same ancestry assignment.

TreeLoom organizes familiar neural operations around these constraints. It passes messages in both directions through every local tree, uses changes in descendant groups to condition a second traversal, and carries a separate memory for the same sampled haplotype across trees. A final gated recurrent unit (GRU) then reads the ordered local representations and predicts ancestry. Sample memory acts inside the tree encoder, whereas the final GRU acts on its outputs.

We evaluate the resulting model on a shared European development endpoint and examine the specific value of added memory with a matched control. The final model has higher measured span accuracy than the evaluated AncestralPaths and ARGMix pipelines. Its gain over the control without added memory is much smaller and accompanies lower boundary F1. The scientific contribution is a genealogy representation and its empirical assessment, with clear limits on what the present comparison can establish. The eight validation simulations have been reused during development, one optimizer seed is available, and an independent test and remaining native comparisons are unfinished.

2 Related Work

LAI from genealogies and genetic variation. AncestralPaths uses genealogical paths to distinguish ancestry routes that can share the same broad source population (Irving-Pease et al. 2024). ARGMix learns graph representations of genealogical relationships together with genomic sequence context (Shanks et al. 2026). ARGformer learns path representations with a transformer and uses a learned embedding for downstream ancestry analysis (Bonet et al. 2026). FLARE estimates local ancestry directly from haplotypes (Browning, Waples, and Browning 2023). These approaches differ in representation, supervision and prediction unit. Our European comparison evaluates the available completed pipelines on a common output endpoint, with the release and task differences stated below.

Neural computation on changing genealogies. A rooted tree provides a natural order for graph message passing. An upward traversal summarizes descendants, and a downward traversal supplies information from the rest of the tree. Genomic recurrence adds a second order, along the chromosome rather than along branches. GRUs provide an established mechanism for combining such sequential context (Cho et al. 2014). TreeLoom uses these standard ingredients to preserve complete local relationships and the correspondence between sampled haplotypes. The relevant empirical questions concern how they are organized for LAI and what added context contributes under matched conditions.

Representations of graph collections. Graph Set Transformer interleaves propagation within member graphs with context from the surrounding collection, allowing information exchange before a final graph readout (Escrig Molina, Chen, and Probst 2026). Temporal graph state space models likewise study recurrent state on changing graphs (Li et al. 2024). These perspectives are relevant to a sequence of local genealogies. The genomic setting additionally supplies an order, unequal physical spans and persistent sampled haplotypes, which constrain the relationships that can carry information between trees. This connection motivates the representation without establishing transfer beyond ancestry inference.

3 TreeLoom

3.1 Prediction Task and Model Overview

Let $T_t = (V_t, E_t)$ be the inferred local tree covering genomic interval I_t, with physical length w_t. Let q index a sampled haplotype and $v_t(q)$ its node in tree t. All sampled haplotypes appear in every tree. The model predicts a probability vector p_{tq} over four source ancestries: Anatolian farmer (Ana), Caucasus hunter-gatherer (CHG), western hunter-gatherer (WHG) and eastern hunter-gatherer (EHG).

The inputs are phased observed variants, inferred tree topology and node ages, sampling ages, and known source labels for reference haplotypes. Query source labels are masked. Simulation ancestry is used only as external training supervision or evaluation truth. It does not define the input features, inferred trees, structural groups or prediction intervals.

Figure 1 shows the complete model. An encoder of observed data first initializes the nodes. A complete upward and downward tree traversal produces local context. Structural conditioning and sample memory modify the node inputs, after which a second complete traversal uses the same learned propagation weights. Its outputs for sampled nodes are aligned by haplotype identity across trees and passed to a final genomic GRU and classifier. No tree is reduced to a single pooled vector before query prediction.

3.2 Observed Evidence and Complete Tree Traversal

For each haplotype, a genetic encoder summarizes its called alleles relative to observed donor and panel allele frequencies. Mean and maximum pooling within a tree interval produce a local genetic representation. This representation is combined with sampling age and the known reference label when available. Inferred age, status as a sampled node and root status provide features for internal nodes. These features initialize a node state b_v. Branch geometry and summaries of agreement between mutations and observed alleles determine message gates.

The upward traversal combines descendant evidence. For a parent v and child u,

$$m_{u\to v} = \frac{g_u^{\uparrow} \odot W_{\uparrow} h_u}{\max(1, d_v)}, \tag{1}$$

$$s_v = \sum_{u \in \mathrm{child}(v)} m_{u\to v}, \tag{2}$$

$$h_v = \mathrm{LN}(b_v + \tanh(A_{\uparrow}[b_v, s_v] + a_{\uparrow})). \tag{3}$$

Here d_v is the number of children, $g_u^{\uparrow}$ is a gate computed from observed features, $\odot$ denotes elementwise multiplication, and brackets denote concatenation. Normalization prevents the summed message magnitude from simply increasing with the number of children.

The downward traversal supplies the context outside a child's branch. For child u, the parent context is

$$c_{v\setminus u} = o_v + b_v + s_v - m_{u\to v}. \tag{4}$$

A gated learned transformation maps this context to the child's outside state o_u. The outside state is zero at the root. Subtracting the recipient's upward contribution prevents that contribution from immediately returning as evidence from elsewhere in the tree. Together, h_v and o_v represent information below and outside a node. Both traversals visit every node, including inferred ancestors.

3.3 Conditioning on Local Tree Structure

We describe structural changes using descendant sets rather than presumed identities of inferred ancestors. Let $C_t(v)$ be the set of sampled haplotypes descended from node v. A proper clade contains at least two sampled haplotypes and excludes the complete panel. When a unary chain has several nodes with the same descendant set, we use its deepest representative.

A node's gained flag g_v is one when its proper clade is absent from the preceding tree. Its lost flag ℓ_v is one when the

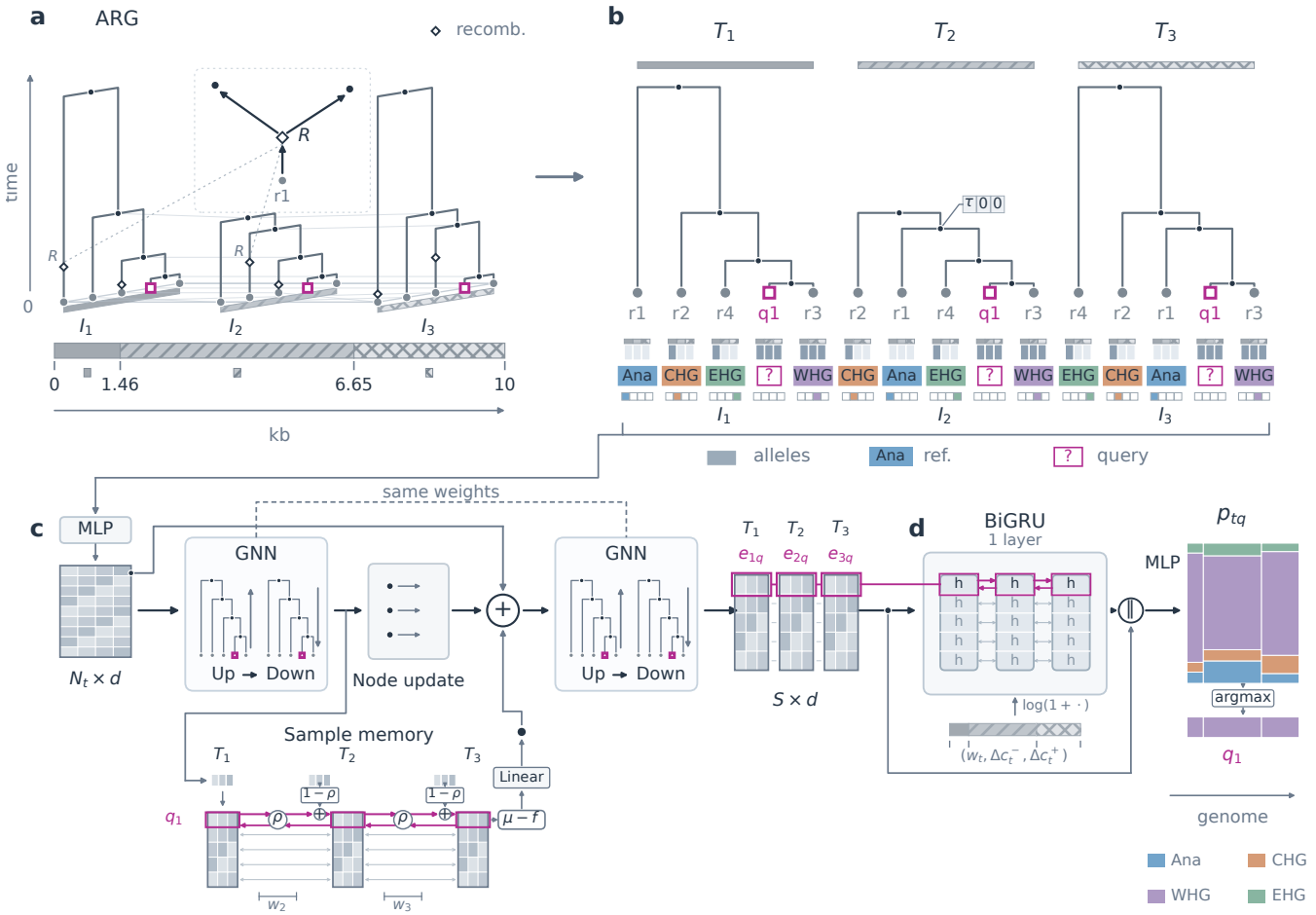


Figure 1: **(a,b)** Three intervals of an illustrative simulated ARG yield matching local trees. All sampled haplotypes supply observed alleles. References supply ancestry vectors, while query ancestry is unknown. Ancestors carry age τ and sampled/root indicators. **(c)** N_t counts nodes, S sampled haplotypes and d feature width. Two GNN passes each traverse every node upward then downward and share weights. Node updates describe changes in descendant groups and each node's state. Sample memory follows persistent sampled identities. In each direction, $\rho_t = \exp(-w_t/\lambda)$ retains history and $1-\rho_t$ weights current group context f. The two histories are averaged. Their current group mean μ is centered by subtracting f, linearly projected and injected before the second GNN pass. All S sampled embeddings remain aligned. e_{tq} embeds haplotype q in tree t. Magenta follows q_1. **(d)** One shared BiGRU with one layer processes each sampled sequence using logarithms of $1 + w_t$ and $1 + \Delta c_t^{\pm}$. These encode span and distances to neighboring interval centers. $\|$ concatenates local and recurrent embeddings. The MLP and softmax give ancestry probabilities p_{tq}, and argmax gives one ancestry call per interval. Feature values and predictions are schematic.

clade is absent from the following tree. Flags at the missing sequence neighbors are zero. The flags condition a local residual,

$$\Delta_v^{\text{change}} = O[g_v(Uh_v \odot Vh_v) - \ell_v(Uh_v \odot Wh_v)]\,. \quad (5)$$

The projections have a lower dimensional intermediate space. An additional local transformation acts on clades that occur in at least two consecutive trees,

$$\Delta_v^{\text{local}} = \chi_v O_l[(U_l h_v) \odot (V_l h_v) + W_l h_v]\,, \quad (6)$$

where χ_v indicates eligibility. This transformation uses the node's own representation. It does not import an embedding from another tree.

These features identify stability and change in the observed topology. Equal descendant sets do not establish that two inferred nodes are the same biological ancestor. The second tree traversal receives $b_v + \Delta_v^{\text{change}} + \Delta_v^{\text{local}}$ together with the contribution from sample memory defined next. It reuses the first traversal's weights and observed gates.

3.4 Memory for the Same Sampled Haplotype

Sample memory means history carried for the same sampled haplotype across genomic trees. It is distinct from the final genomic GRU. This memory modifies the internal tree representation before the second traversal.

We cut each inferred tree at a fixed age threshold and group sampled haplotypes by their ancestor at the cut. Samples older than the cut retain their own sampled node. The cut uses inferred node ages and sampling ages, without query ancestry labels. For group k in tree t, a learned projection of the sum of upward and outside states at its representative gives a context vector f_{tk}. Let $k_t(q)$ be haplotype q's group.

A forward scan carries a state for each haplotype,

$$z^{\rightarrow}_{tq} = \rho_t z^{\rightarrow}_{t-1,q} + (1-\rho_t) f_{t,k_t(q)}, \qquad \rho_t = \exp(-w_t/\lambda). \tag{7}$$

The first state equals its current context vector. A reverse scan applies the same rule in reverse genomic order. The learned positive length scale λ determines how much history is retained over a given physical span. A longer interval places more weight on its current context.

We average the two scan directions, average the resulting haplotype states within each current group, and subtract the group's current context vector. A learned output projection converts this difference to a residual at the group representative. The second tree traversal propagates that residual through the complete genealogy. Sample identity remains the link across trees even when the inferred groups change.

The matched control, TreeLoom-NoMem, omits this added memory and retains both complete traversals, local structural conditioning and the final genomic GRU. The phrase *without memory* therefore refers only to the additional mechanism in Equation 7.

3.5 Genomic Context and Ancestry Prediction

After the second traversal, a learned projection of the upward and outside states produces a local embedding $e_{tq} \in \mathbb{R}^d$ for every sampled haplotype. Alignment across trees is a direct lookup by the same sampled identity. It does not require matching internal ancestors.

A bidirectional GRU with one layer processes the local embeddings together with interval geometry,

$$r_{1:T,q} = \mathrm{BiGRU}\left([e_{tq}, \mathrm{geometry}(I_t)]_{t=1}^{T}\right). \tag{8}$$

The geometry contains interval length and distances to neighboring interval centers. The GRU has hidden width h in each direction. A classifier combines the local embedding of width d with the resulting genomic representation of width $2h$,

$$p_{tq} = \mathrm{softmax}\left(\mathrm{MLP}([e_{tq}, r_{tq}])\right). \tag{9}$$

All sampled haplotypes and tree intervals remain present during these computations. Query selection occurs only at readout. Predictions are constant within an inferred tree interval, so possible ancestry transitions are restricted to its edges.

4 Evaluation Design

4.1 European Development Simulations

The completed European comparison uses 32 training simulations and eight validation simulations, each 2 Mb long. The demographic formulation is adapted from the released AncestralPaths workflow associated with Irving-Pease et al. (2024). The panel contains 476 diploid individuals, or 952 phased haplotypes. There are 266 reference and 686 query haplotypes. Queries comprise modern individuals, Bronze Age individuals, Bronze Age Anatolians (BAA), Neolithic farmers (Neo) and Yamnaya (Yam). Simulation identity determines the split before tree intervals, rows or sequence chunks are constructed.

Relate infers genealogies from the observed variants and sampling ages (Speidel et al. 2019). All completed European pipelines use the common observed panel and inferred genealogies, represented as required by each method. The simulation assumptions and inference parameters are in the supplement. The released Bronze Age roster is only an approximate correspondence to the ARGMix preprint's Post-Neolithic category.

The eight validation simulations have informed repeated model development and checkpoint selection. We therefore describe all results on them as development evidence. Only one optimizer seed is available for each evaluated condition. Intervals, bins and haplotypes from a simulation are dependent observations, not independent biological replicates.

4.2 Common Scoring Endpoint

The primary endpoint is accuracy weighted by physical span for the 182 modern query haplotypes. Secondary endpoints cover the ancient cohorts and all queries. All European predictions are scored on the same fixed 100 bp bins clipped to inferred coverage. Let m_{iqc} be the physical mass of true ancestry c for query q in bin i. Span accuracy is

$$A = \frac{\sum_{iqc} m_{iqc}\, \mathbf{1}\{c = \arg\max_k p_{iqk}\}}{\sum_{iqc} m_{iqc}}. \tag{10}$$

Each simulation receives equal weight in the reported mean. We give sample SD across the eight simulations in percentage points. This SD describes simulation variation for a fitted model. It is neither uncertainty across training seeds nor a confidence interval.

Accuracy measures the amount of correctly assigned sequence. Boundary precision, recall and F1 assess whether the ancestry transitions are located correctly. Boundaries are matched one to one within 1 kb. The matcher uses position and does not require the predicted transition to have the correct pair of source labels. Negative log likelihood (NLL) uses the same physical masses and the original predicted probabilities, without flooring or recalibration.

4.3 Comparator Scope and Matched Memory Control

The AncestralPaths comparison uses the released dense six output classifier. We sum its six path probabilities into the four source probabilities before class assignment,

$$(p_{\mathrm{Ana}}, p_{\mathrm{CHG}}, p_{\mathrm{WHG}}, p_{\mathrm{EHG}}) = (p_1 + p_5, p_2 + p_6, p_3, p_4). \tag{11}$$

This derived endpoint differs from the native six path task. The paper describes a convolutional classifier, while the executed archive contains a dense classifier. Its bounded replacement sampler also leaves 9,126 requested BAA path six

training rows unfilled, with an unmeasured effect on accuracy. The five epoch author schedule is completed and its final epoch is used.

ARGMix uses the current author release, with 15 references and no ancestor inputs. The preprint describes 25 references and three inferred ancestors. The public release does not provide a supported implementation of that earlier input configuration. A panel slot and metadata adaptation allows the released model to process the common roster. The model, training augmentation and native rule for stopping are retained. Eleven epochs complete, and its native validation objective selects epoch nine.

TreeLoom also receives observed SNP features. Methods differ in reference eligibility, representation, supervision unit, effective training diversity, cumulative optimization and checkpoint selection. The common scorer makes their outputs comparable on this endpoint, but does not make their information or training budgets equivalent. Completing an author schedule also does not establish convergence. These are named pipeline comparisons, with no exact publication reproduction or overall LAI ranking established.

The memory comparison asks a narrower question. TreeLoom and TreeLoom-NoMem start from the same trained local encoder, receive the same additional 960 updates and keep all weights trainable. They use the same checkpoint rule. The memory model has 513 additional parameters. Each arm selects its checkpoint independently. Earlier developmental models received different cumulative training exposure and are not treated as component ablations.

Scientific TODO. Resolve the earlier ARGMix model, reference roster, genomic map, genotype error settings and evaluation protocol before any claim of exact publication reproduction. Recover AncestralPaths' stated convolutional model and original evaluator, and assess the realized sampling deficit.

5 Results

5.1 Span Accuracy on the Shared European Endpoint

TreeLoom reaches $97.08 \pm 0.57\%$ modern span accuracy (Table 1). The evaluated current ARGMix release reaches $94.45 \pm 1.15\%$, and the released dense AncestralPaths classifier reaches $93.78 \pm 0.84\%$ on the derived four source endpoint. TreeLoom assigns 2.63 percentage points more sequence correctly than this ARGMix condition.

This result supports the fitted TreeLoom pipeline on the evaluated data. It does not isolate the effect of propagation through complete trees or added sample memory from all other pipeline differences. The corresponding matched memory comparison is much closer.

5.2 Accuracy across Modern and Ancient Cohorts

TreeLoom has the highest measured accuracy in each listed cohort (Table 2). Differences among cohorts reflect their distinct simulated histories. Small cohorts such as BAA and Yam provide limited evidence about individual ancestry transitions.

Pipeline	Accuracy ↑	Boundary F1 ↑
AncestralPaths	93.78 ± 0.84	7.03 ± 0.78
ARGMix, current release	94.45 ± 1.15	11.74 ± 1.73
TreeLoom	$\mathbf{97.08 \pm 0.57}$	$\mathbf{18.38 \pm 1.31}$

Table 1: Modern European development performance, mean±sample SD in percent across eight 2 Mb simulations and 182 query haplotypes. AncestralPaths probabilities are collapsed from six paths to four source classes. ARGMix uses the current author release. Boundary matching uses a 1 kb positional tolerance. Bold marks the highest mean. Inputs, training and selection differ between pipelines as detailed in the text.

5.3 Added Memory Changes Accuracy and Boundaries Differently

TreeLoom-NoMem reaches $97.00 \pm 0.50\%$ accuracy, only 0.08 percentage points below TreeLoom (Table 3). Its boundary F1 is higher, at $19.05 \pm 1.43\%$ compared with TreeLoom's $18.38 \pm 1.31\%$. The paired mean changes after adding sample memory are therefore $+0.08$ accuracy points and -0.67 boundary F1 points.

The practical implication depends on the intended use. More correctly assigned genomic mass can help estimate broad ancestry patterns, while a fragmented or displaced set of transitions can distort the reconstructed tracts. The memory mechanism improves the former metric slightly in this condition and reduces the latter. The final model name denotes the highest observed condition by span accuracy, rather than a model that wins every endpoint.

Additional matched controls give little evidence that more elaborate exchange between trees solves this problem. Replacing local node conditioning with context from other occurrences of the same clade changes accuracy by only $+0.01 \pm 0.15$ points. Alternative transport rules for memory also fail the registered practical criteria. These controls are reported in the supplement without conflating them with the differently trained developmental stages.

5.4 High Span Accuracy Leaves Substantial Tract Error

Boundary F1 remains low for all three completed pipelines. TreeLoom reaches 18.38%, despite assigning 97.08% of the represented genomic span correctly. This gap shows that a small amount of incorrectly assigned sequence can coexist with many missed or extra ancestry transitions. Span accuracy alone is therefore insufficient evidence that ancestry tracts are recovered faithfully.

The model's readout places another constraint on tract recovery: it can change ancestry only at inferred tree edges. True ancestry changes need not occur at these edges or be resolved accurately by the inferred genealogy. Better boundary recovery may require both a more flexible readout and a training objective that rewards transition fidelity. The present results do not establish the effect of either change.

Scientific TODO. Complete a matched American comparison on common observations, reference and query identities,

Cohort	Query haplotypes	AncestralPaths	ARGMix	TreeLoom
Modern	182	93.78 ± 0.84	94.45 ± 1.15	$\mathbf{97.08 \pm 0.57}$
Bronze Age	162	94.76 ± 1.18	97.71 ± 0.75	$\mathbf{99.16 \pm 0.43}$
BAA	14	96.94 ± 1.32	98.44 ± 1.02	$\mathbf{99.35 \pm 1.04}$
Neo	302	97.57 ± 0.27	99.30 ± 0.26	$\mathbf{99.73 \pm 0.07}$
Yam	26	95.43 ± 1.41	99.22 ± 0.90	$\mathbf{99.94 \pm 0.12}$
Ancient	504	96.54 ± 0.32	98.76 ± 0.31	$\mathbf{99.55 \pm 0.13}$
All queries	686	95.81 ± 0.45	97.61 ± 0.50	$\mathbf{98.89 \pm 0.23}$

Table 2: Cohort span accuracy in percent, mean±sample SD across the same eight development simulations. AncestralPaths uses the derived four source endpoint, and ARGMix uses the current author release. Bold marks the highest mean per row. Ancient and all queries aggregate the corresponding cohorts above them. The Bronze Age roster is not established as identical to the ARGMix paper's Post-Neolithic panel. Cohorts differ in sampling age, ancestry mixture and tract structure, so high simulated ancient accuracy does not establish robustness to empirical ancient DNA.

Matched model	Accuracy ↑	Boundary F1 ↑
TreeLoom-NoMem	97.00 ± 0.50	$\mathbf{19.05 \pm 1.43}$
TreeLoom	$\mathbf{97.08 \pm 0.57}$	18.38 ± 1.31

Table 3: Effect of added sample memory after the same initialization and additional training budget. Values are mean±simulation SD in percent on the same modern endpoint. NoMem retains the final genomic GRU and both complete tree traversals. Checkpoints are selected separately within each arm. Bold identifies the higher mean per metric.

focal records and truth support. Evaluate independent simulations, additional optimizer seeds and larger European panels. Resolve the American demographic and ploidy reconstruction and ARGformer's published downstream evaluator before drawing broader comparative conclusions.

6 Discussion

TreeLoom connects two useful views of ancestry. Within a local tree, reference and query genomes are related through shared ancestors. Along the genome, the same sampled haplotype has a continuous sequence of local relationships. Complete tree traversal makes both descendant and outside context available at each sampled node, and genomic recurrence combines that context in physical order. Sample memory provides an additional route for history to influence the tree representation without matching inferred internal ancestors.

The completed comparison is encouraging for this representation, but its strongest claim is descriptive. TreeLoom performs well on eight reused development simulations under a specified demographic formulation. The native comparator pipelines preserve available author algorithms while differing materially from the corresponding publication recipes and from TreeLoom's input and training choices. The measured gap cannot be assigned entirely to a particular neural component. Independent simulations, multiple training seeds and faithful completion of the native comparisons are needed to assess generalization and relative method strength.

The matched evidence is more specific. Added sample memory brings a small mean accuracy gain and reduces boundary F1. Context from the same clade in other trees and more elaborate memory transport also yield limited gains under their matched controls. These findings favor a restrained interpretation of interaction between trees. They do not establish a new generally superior operator, and they suggest that tract boundaries deserve direct attention during model design and evaluation.

The simulated ancient cohorts are also an incomplete model of ancient genomic data. Their sampling history is explicit, but uncertainty in phasing, genotype calls, reference representation, demographic assumptions and genealogical reconstruction can change the information available to LAI. A future empirical validation needs evidence that is independent of the model's own ancestry assignments. A descriptive application of an earlier TreeLoom condition and its counterevidence are retained in the supplement, without treating model output as known mixture fractions.

TreeLoom provides a concrete framework for learning LAI from complete local genealogies. Its present results motivate further study of that representation while identifying a central limitation: assigning most bases correctly does not ensure that ancestry tracts have been reconstructed correctly.

Code availability. Source code, experiment configurations and complete results are available at https://github.com/alexeyshmelev/ARG-GNN.

References

Baumdicker, F.; Bisschop, G.; Goldstein, D.; et al. 2022. Efficient ancestry and mutation simulation with msprime 1.0. *Genetics*, 220(3).

Bonet, D.; Shanks, C.; Comajoan Cara, M.; Abante, J.; and Ioannidis, A. G. 2026. ARGformer: learning on ancestral recombination graphs with transformers. *Bioinformatics*, 42(Supplement 2): btag438.

Browning, S. R.; Waples, R. K.; and Browning, B. L. 2023. Fast, accurate local ancestry estimation with FLARE. *The American Journal of Human Genetics*, 110(2): 326–335.

Cho, K.; van Merrienboer, B.; Gulcehre, C.; Bahdanau, D.; Bougares, F.; Schwenk, H.; and Bengio, Y. 2014. Learning Phrase Representations using RNN Encoder–Decoder for Statistical Machine Translation. In *Proceedings of the 2014*

Conference on Empirical Methods in Natural Language Processing, 1724–1734.

Escrig Molina, J. E.; Chen, B.; and Probst, D. 2026. Graph Set Transformer. arXiv:2606.05116.

Irving-Pease, E. K.; Refoyo-Martínez, A.; Barrie, W.; Ingason, A.; Pearson, A.; Fischer, A.; Sjögren, K.-G.; Halgren, A. S.; Macleod, R.; Demeter, F.; Henriksen, R. A.; Vimala, T.; McColl, H.; Vaughn, A. H.; Speidel, L.; Stern, A. J.; Scorrano, G.; Ramsøe, A.; Schork, A. J.; Rosengren, A.; Zhao, L.; Kristiansen, K.; Iversen, A. K. N.; Fugger, L.; Sudmant, P. H.; Lawson, D. J.; Durbin, R.; Korneliussen, T.; Werge, T.; Allentoft, M. E.; Sikora, M.; Nielsen, R.; Racimo, F.; and Willerslev, E. 2024. The selection landscape and genetic legacy of ancient Eurasians. *Nature*, 625(7994): 312–320.

Li, J.; Wu, R.; Jin, X.; Ma, B.; Chen, L.; and Zheng, Z. 2024. State Space Models on Temporal Graphs: A First-Principles Study. In *Advances in Neural Information Processing Systems*, volume 37, 127030–127058.

Shanks, C.; Bonet, D.; Comajoan Cara, M.; and Ioannidis, A. G. 2026. Graph transformer for ancient ancestry inference. *bioRxiv*. Preprint, version 1, posted 7 April 2026.

Speidel, L.; Forest, M.; Shi, S.; and Myers, S. R. 2019. A method for genome-wide genealogy estimation for thousands of samples. *Nature Genetics*, 51(9): 1321–1329.

Supplementary Material

A Simulation Assumptions and Parameters

The European simulations adopt the released AncestralPaths demographic formulation associated with Irving-Pease et al. (2024). The numerical assumptions below define this study condition. They are not demographic estimates inferred from the simulated observations. A census at 350 generations defines four source ancestry supervision before the older mergers of source populations. This truth census is distinct from the model's observed tree cut at the same nominal time.

The panel has 343 query and 133 diploid individuals from pure reference populations. Queries comprise 91 modern, 81 Bronze Age, seven BAA, 151 Neo and 13 Yam diploids. Pure references comprise 51 WHG, 43 EHG, 25 Ana and 14 CHG diploids. Modern samples have age zero, and ancient samples retain the released age roster in generations. The resulting 952 input haplotypes include 686 queries.

Time (generations)	Descendant	Ancestral proportions
166	Present/Bronze	0.5 Neo, 0.5 Yam
169	BAA	0.25 CHG, 0.75 Ana
177	Yam	0.5 CHG, 0.5 EHG
259	Neo	0.25 WHG, 0.75 Ana
600	WHG, EHG	Shared NE ancestor
800	CHG, Ana	Shared WA ancestor
1,500	NE, WA	Common trunk

Table 4: European demographic events adopted from the released formulation associated with AncestralPaths (Irving-Pease et al. 2024). Times are measured backward from the present. Proportions specify ancestral contributions at the corresponding event.

Effective population sizes are 50,000 for Present/Bronze and Neo, 20,000 for Yam and BAA, 3,000 for EHG and CHG, 2,000 for WHG, and 5,000 for Ana. The older NE, WA and trunk populations each have size 15,000. Growth and migration between events are zero. The simulation uses the Hudson coalescent, discrete genomic positions, a uniform recombination rate of 10^{-8} per base per generation, and JC69 mutations at 1.25×10^{-8} per base per generation (Baumdicker et al. 2022). The 2 Mb sequence length and 32/8 training/validation split define the present development panel.

Relate uses the observed biallelic variants and sampling ages (Speidel et al. 2019). Its initial effective haploid size is 30,000, mutation rate is 1.25×10^{-8} and generation time is 28 years. The European inference includes five iterations of population size estimation followed by estimation of branch lengths. All applicable completed model adapters consume this same final reconstruction. Mutation agreement features use observed alleles on the inferred trees.

Scientific TODO. Verify and cite the exact primary source location of each demographic assumption and the complete ancient sampling age roster. Add the actual simulation seed list, software versions, genotype error condition and genomic coordinate convention. Resolve differences from the ARGMix paper's empirical map, error model, demographic perturbations and eligible reference ages before calling the condition a reproduction. For the American benchmark, provide a primary source demographic table and an explicit account of the haplotype interpretation and focal coordinate assumptions.

B Model and Training Parameters

The genetic site encoder has two hidden layers of width 32. For a called allele it receives donor log frequency features, corresponding availability indicators, panel frequency, site entropy and a call indicator. Eight donor groups combine the four reference ancestries with ages below or at least 350 generations. Allele frequencies use additive pseudocount 0.5. If a represented reference class has no called alleles at a site in an age group, all represented classes in that age group are masked for that site. Mean and maximum pooling are combined with call count, missing pool status and interval span.

The inferred tree cut for sample memory is fixed at 350 generations. Samples older than the cut retain their own sampled node. Tree states have width 32 and local residual projections have rank eight. The memory projection also has rank eight and adds 513 parameters. Its length scale starts at 5 kb and is constrained to 100 bp–100 kb. Residual output projections start at zero. The genomic GRU has one layer and width 32 in each direction. The ancestry classifier receives 96 features, maps through width 32 and produces four scores. TreeLoom has 32,325 parameters. TreeLoom-NoMem has 31,812.

Training minimizes physical span cross entropy over all 686 queries,

$$\mathcal{L} = -\frac{\sum_{tqc} M_{tqc} \log p_{tqc}}{\sum_{tqc} M_{tqc}}, \qquad M_{tqc} = \sum_{i:r(i)=t} m_{iqc}. \tag{12}$$

Here $r(i)$ maps a scored bin to its inferred tree interval. Summing target masses within an interval preserves the original objective because the model's prediction is constant there. A complete simulated chromosome is one optimization example.

AdamW uses learning rate 10^{-3}, weight decay 10^{-4}, betas $(0.9, 0.999)$ and epsilon 10^{-8}. Gradient norm is clipped at one. The reported seed is 101. The shared initializer is trained over three successive 30-pass stages on the 32 training simulations, totaling 2,880 updates as tree processing and local conditioning are introduced. These successive stages are a training lineage, not matched component ablations. Optimizer moments reset at each stage.

For the final comparison, both TreeLoom and NoMem start from that same 2,880-update model and complete 30 further passes, or 960 new updates. All inherited weights remain trainable. Checkpoints every five passes are ranked by equal simulation modern validation accuracy, then lower NLL, then earlier pass. TreeLoom selects added pass 30, or 3,840 cumulative updates. NoMem selects added pass 20, or 3,520 cumulative updates, while its run also completes 3,840 updates. The alternate memory rules below share the same initializer, capacity and completed budget as TreeLoom.

C Comparator Conditions and Unresolved Recipe Differences

AncestralPaths. The executed archive contains a flattening operation on a 5×9 representation, a dense layer of width 512 with ReLU, dropout 0.2 and six softmax outputs. This differs from the convolutional classifier described in the paper (Irving-Pease et al. 2024). The fit completes five epochs with Adam learning rate 0.001, batch size 32 and shuffling. A deterministic seed of 101 is supplied by the adaptation. The final epoch is used without validation checkpoint selection. Modern deterministic compatibility does not establish numerical equivalence to the original legacy execution.

The bounded sampler accepts 5,190,874 of 5,200,000 requested rows with replacement. All 9,126 missing rows are BAA path six examples, amounting to 2.2815% of requested BAA rows and 0.1755% of all requested rows. There are 53,214 replacement duplicates. Other cohort quotas are met. The sampler stops after no progress or a bound of 128 passes, so its realized shortfall is a material adaptation whose accuracy effect has not been measured. Genomic training diversity is 32×2 Mb here, whereas the paper describes 3×200 Mb. Completing five epochs does not establish convergence, and training loss is still decreasing at epoch five.

The four source probability collapse in the main text is separate from six path prediction, a retrained four path task, or evaluation restricted to non BAA routes. It must not be compared directly with a published native six path accuracy. Strict NLL is infinite in the present derived endpoint because a true source receives zero probability over 800 query base pairs. No probability floor is applied.

ARGMix. The evaluated author release uses 15 references without inferred ancestor nodes. Its transformer, bidirectional GRU and native training procedure are retained (Shanks et al. 2026). It uses 1,024-row chunks with stride 512, yielding 21,952 training and 5,488 validation chunks. All 686 queries are available, with no local subgraph cap. Training drops the nearest reference, validation drops none, label smoothing is 0.1 and loss weighting is absent. Source seed 50 is retained. The model uses supervision at the first SNP of an interval and averages overlapping softmax predictions arithmetically.

Its native selection objective is the smoothed, masked and unweighted mean validation batch loss. The selected epoch is nine of 11 completed epochs, with patience two and a 20-epoch cap. The external span scorer does not replace this inner selection rule. Its preprint's 25-reference and model with three ancestor inputs is not recovered by the current release, whose supported loaders and extraction reject ancestor inputs. A current release run with 25 references would still have zero ancestors and would be a separate sensitivity condition. The original model, exact reference eligibility, checkpoint and publication evaluator remain unresolved. Differences in the observed panel, map, error process, training diversity and prediction support also remain material.

American comparators. A separate American benchmark uses a 30 Mb simulation with 600 reference and 1,000 query haplotypes from EUR, EAS and AFR sources. Native FLARE calls score 98.0100% accuracy and 97.7151% macro F1 on 10,000 predetermined focal records from one validation simulation. This metric counts focal records rather than physical span and has no between simulation SD. It therefore cannot be compared numerically with the European span endpoint or establish a comparative ranking on its own.

The completed FLARE condition uses author release v0.6.0 with native defaults: minimum minor allele count 50, minimum minor allele frequency 0.005, expectation maximization enabled, ten generations, ancestry proportion updates disabled, array mode disabled and seed -99999. Native filtering retains 27,558 of 228,477 observed markers. The evaluator uses original hard ancestry calls at the nearest native marker, with midpoint boundaries, on all 10,000 fixed focal records. It does not score a filtered subset chosen after the outcome. All prediction identities and coordinates are checked before private truth is opened. This focal score from one simulation supplies no estimate of between parent variation.

ARGformer's completed American ancestry pipeline remains unavailable. Its published PCA/clustering classifier and nearest reference retrieval are distinct downstream routes (Bonet et al. 2026). Exact clustering, class mapping and American retrieval settings are not recovered. The prepared route retains the author release's learned normalized embedding and nearest reference retrieval of the nearest reference as a disclosed reconstruction. The paper states contrastive temperature 0.05 and inverse frequency anchor weighting, whereas the release configuration uses 0.30 and its stock route can use uniform weights. These differences remain unresolved. Partial pretraining does not supply ancestry accuracy.

Scientific TODO. Before a claim of publication reproduction, recover the missing original model and evaluator details rather than silently substituting plausible choices. Assess sensitivity to the AncestralPaths sampling deficit, ARGMix reference restriction and ARGformer objective differences where conclusions depend on them.

D Additional Scientific Results

D.1 Limited Matched Tests of Additional Context

A matched test replaces the own node context in Equation 6 with a span weighted average from the other trees in the same consecutive proper clade episode. For member i of episode $E(i)$,

$$\bar{h}_{-i} = \frac{\sum_{j \in E(i)} w_j h_j - w_i h_i}{\sum_{j \in E(i)} w_j - w_i}. \tag{13}$$

Only episodes with at least two trees are eligible. Replacing $V_l h_i$ with $V_l \bar{h}_{-i}$ leaves the parameter count, eligible nodes, initialization and added training budget unchanged. This comparison tests the value of information from other trees beyond an extra local transformation.

The paired accuracy difference is 0.0096 ± 0.1527 percentage points. The paired boundary F1 difference is 0.1001 ± 0.9632 points. The accuracy gain falls below the registered 0.10-point practical threshold.

We also compare sample memory with two alternatives of the same capacity. Group transport pools history through

Conditioning context	Accuracy ↑	Boundary F1 ↑
Own node	96.82 ± 0.61	18.06 ± 1.78
Other episode members	$\mathbf{96.83 \pm 0.59}$	$\mathbf{18.16 \pm 1.38}$

Table 5: Matched test of context from other trees at an earlier shared initialization, with 960 added updates per arm. Values are mean±simulation SD in percent on modern queries. Both arms select pass 30. This is a conceptual context comparison, not a comparison of final training budgets with TreeLoom.

Memory rule	Accuracy ↑	Boundary F1 ↑
Group transport	97.05 ± 0.58	18.85 ± 1.05
Intersection transport	97.06 ± 0.56	18.93 ± 1.42
Same sampled haplotype	$\mathbf{97.08 \pm 0.57}$	18.38 ± 1.31
No added memory	97.00 ± 0.50	$\mathbf{19.05 \pm 1.43}$

Table 6: Matched memory rules, with the same initialization and 960 added updates per arm. Values are mean±simulation SD in percent. The first three arms each have 32,325 parameters and select added pass 30. The control has 31,812 parameters and selects pass 20. Bold identifies the largest mean per metric.

the intermediate cut group when group membership changes. Intersection transport retains the intersections of consecutive groups as the units of transport. Both use the same context from the first traversal, distance weighting, second traversal and final genomic readout as sample memory.

Intersection transport changes accuracy by $+0.0137 \pm 0.0243$ points relative to group transport and by -0.0138 ± 0.0384 points relative to sample memory. It fails both registered practical criteria against every control: at least 0.10 accuracy points gained without mean boundary F1 loss, or at least one boundary F1 point gained with at most 0.05 accuracy points lost. These are development utility thresholds, not statistical significance tests.

D.2 Descriptive Application to Ancient Danish Genomes

An earlier TreeLoom condition with context from other members of a clade episode was applied without further fitting to ten ancient Danish query individuals and eight proxy reference individuals from the imputed genomes analyzed by Irving-Pease et al. (2024). This is the episode condition in Table 5, not the final model with sample memory. Five earlier individuals date to 5.44–5.67 ka cal BP and five later individuals to 3.97–4.26 ka cal BP. Three fixed 10 Mb windows on chromosomes 20–22 contain 1,158 inferred trees and 8,587 qualified site records.

Assignments weighted equally across people integrated over physical span are 14.27 percentage points higher for EHG and 5.52 points higher for CHG in the later group. EHG has a positive contrast on each chromosome. CHG depends on chromosome 21 and changes to -0.68 points when it is omitted. Separate genotype affinity contrasts from the same imputation resource do not support the registered directions. There is no local ancestry truth or calibration of the model's outputs as mixture fractions. These descriptive predictions therefore illustrate application to empirical genealogies and do not validate the historical interpretation.